MODULE 3: READING
(CO: 4)

Complete Study Textbook, Professional Writing Guide & VIT FAT Question Bank

Aligned with Official VIT BAENG101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 10 Topics): (1) Analysing definitions: Types of definitions, (2) Methods for Expanding Definitions, (3) Reading Comprehension: Drawing Inferences from Scientific Texts, (4) Critical reading strategies, (5) Structure of a research article (IMRAD), and (6) Analysing and dissecting a published research article.

1 VIT FAT Exam Blueprint & Master Rules

Module 3 carries **15 to 20 marks** in BAENG101 FAT:

1. **Formal & Expanded Technical Definitions (10 Marks):** Writing formal three-part definitions (Term = Class / Genus + Differentia) and expanding them using etymology, operating principle, physical components, and exemplification.
2. **Reading Comprehension & Critical Inferences (10 Marks):** Analyzing an unfamiliar scientific passage, distinguishing explicit facts from subtle author inferences, and mapping the IMRAD research architecture.

Definition Formula	Components	Worked Example
Formal Definition	Term = Class + Differentia	A **microcontroller** (Term) is an **integrated circuit** (Class) that **contains a processor core, memory, and programmable I/O peripherals on a single chip to control embedded systems** (Differentia).
Expanded Definition Methods	Explication, Operating Principle, Contrast, Analogy	Expanding 'Supercapacitor': Describe double-layer electrostatic charge storage vs electrochemical battery intercalation.

2 Structure of a Scientific Research Article (IMRAD)

2.1 The IMRAD Framework

1. **Introduction (Why was the study undertaken?):** Identifies the overarching research problem, reviews current state-of-the-art literature, highlights the critical research gap, and states the core research objective or hypothesis.
2. **Methods (How was the study conducted?):** Comprehensive, reproducible description of experimental setups, materials, sample synthesis, fabrication techniques, and analytical instrumentation.
3. **Results (What was found?):** Presentation of empirical data using high-resolution graphs, tables, and statistical significance tests without subjective speculation.
4. **Discussion (What do the findings mean?):** Interprets results in relation to the initial hypothesis, compares with previous literature, addresses limitations, and outlines future engineering applications.